

**22AI503 – MACHINE LEARNING****UNIT 1 – INTRODUCTION****TOPIC 6: Regression - Multivariate in Prediction****6.1 Multivariate in Prediction**

Multivariate Prediction is a machine learning technique used to predict an outcome by considering multiple independent variables (features) simultaneously. Instead of relying on a single factor, it analyzes the combined effect of several variables on the target variable.

For example, predicting a student's exam score based on:

- Study Hours
- Attendance Percentage
- Assignment Marks

All three variables together influence the final score.

Definition

Multivariate Prediction is the process of predicting a dependent variable using two or more independent variables.

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

Where:

- Y = Predicted value (dependent variable)
- b_0 = Intercept
- $b_1, b_2, \dots, b_n$ = Regression coefficients
- $X_1, X_2, \dots, X_n$ = Independent variables

Example

A student's score depends on:

- X_1 = Study Hours
- X_2 = Attendance (%)
- X_3 = Assignment Score

Prediction model:

$$\text{Score} = 10 + 5X_1 + 0.3X_2 + 0.8X_3$$

This model uses all three factors to predict the final score.



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited by NAAC with "A+" Grade

SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA

Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

6.2 Importance of Multivariate Prediction

1. Improves prediction accuracy.
2. Captures real-world complexity.
3. Considers multiple influencing factors.
4. Reduces prediction bias.
5. Supports better decision-making.

6.3 Applications

Education

Predict student performance using:

- Attendance
- Study Hours
- Assignment Marks

Healthcare

Predict disease risk using:

- Age
- Weight
- Blood Pressure
- Cholesterol Level

Real Estate

Predict house prices using:

- Area
- Number of Bedrooms
- Location
- Age of House

Finance

Predict credit risk using:

- Income
- Loan Amount
- Credit Score
- Employment History

Sales Forecasting

Predict sales using:

- Advertising Budget



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited by NAAC with "A+" Grade

SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA

Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

- Product Price
- Seasonal Demand

6.4 Advantages

- Higher prediction accuracy
- Better understanding of variable influence
- Handles complex real-world problems
- Supports data-driven decisions
- More reliable predictions

6.5 Challenges

- Multicollinearity among variables
- Increased computational complexity
- Feature selection difficulty
- Risk of overfitting
- Need for larger datasets

6.6 Solved Problem

A company predicts sales using:

$$Y = 500 + 10X_1 + 5X_2$$

Where:

- X_1 = Advertising Budget (₹1000)
- X_2 = Number of Salespersons

Given:

- $X_1 = 20$
- $X_2 = 15$

Solution:

$$Y = 500 + 10(20) + 5(15)$$

$$Y = 500 + 200 + 75$$

$$Y = 775$$

Predicted Sales = 775 units



Discussion Questions

1. Why is multivariate prediction more effective than using a single variable for prediction?
2. How does multicollinearity affect multivariate prediction models?
3. What factors should be considered when selecting variables for a multivariate model?
4. How can multivariate prediction improve business decision-making?
5. What are the challenges of building accurate multivariate prediction models?
6. A model is given by:
 - $Y = 100 + 4X_1 + 2X_2$
 - $X_1 = 10, X_2 = 15$
 - Find Y.
7. A house price model is:
 - a. $\text{Price} = 200000 + 1000X_1 + 50000X_2$
 - b. $X_1 = \text{Area (1200 sq.ft)}$
 - c. $X_2 = \text{Bedrooms (3)}$
 - d. Find the house price.
8. A student's score model is:
 - a. $\text{Score} = 20 + 5X_1 + 0.2X_2$
 - b. $X_1 = \text{Study Hours (8)}$
 - c. $X_2 = \text{Attendance (90\%)}$
 - d. Find the predicted score.
9. An employee salary model is:
 - a. $\text{Salary} = 15000 + 2500X_1 + 1000X_2$
 - b. $X_1 = \text{Experience (5 years)}$
 - c. $X_2 = \text{Certifications (2)}$
 - d. Find the salary.
10. A sales prediction model is:
 - a. $\text{Sales} = 300 + 8X_1 + 4X_2$
 - b. $X_1 = \text{Advertising Budget (25)}$
 - c. $X_2 = \text{Salespersons (10)}$
 - d. Find the predicted sales.